

AS-REP Roasting Exposure Detection & Investigation

PowerShell + Windows Security logs analyzed in Splunk Enterprise

Splunk Enterprise

PowerShell 4104

Windows Security 4738

MITRE T1558.004

Case summary

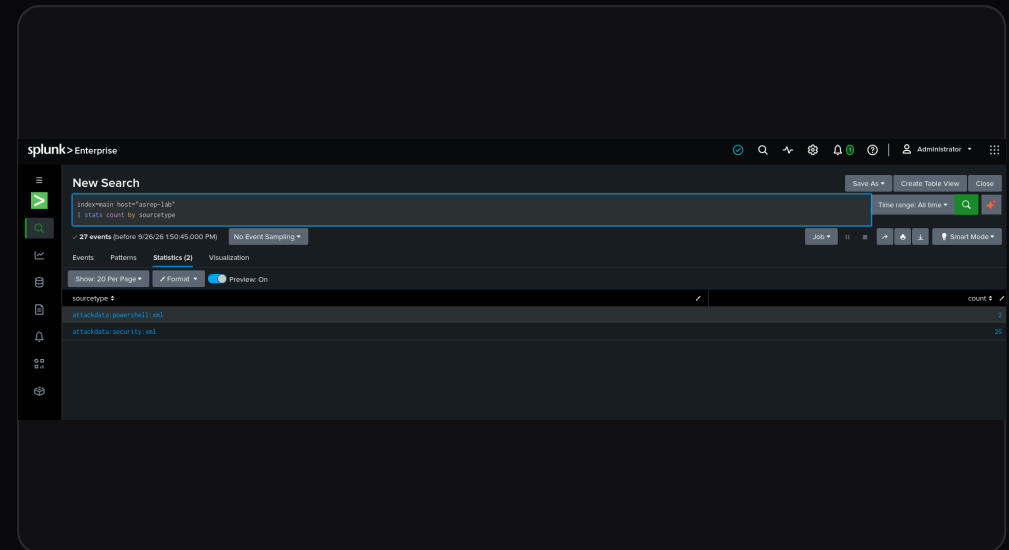
This case study uses official Splunk Attack Data to investigate a PowerShell change to Kerberos pre-authentication, review related Windows account-change events, and build a focused Splunk detection. The report separates what the logs confirm from what they do not prove.

DATA SOURCE Splunk Attack Data - T1558.004 / PowerShell

What was analyzed

Dataset in Splunk

- 2 PowerShell Operational events (Event ID 4104).
- 25 Windows Security events (Event ID 4738).
- The dataset is published by Splunk Attack Data for MITRE ATT&CK T1558.004.
- Goal: identify the risky account-control change and create a reusable detection.



Evidence standard used in this report

- Confirmed: PowerShell logs contain Set-ADAccountControl with -DoesNotRequirePreAuth:\$true.
- Confirmed: Windows Security data contains Event ID 4738 account-change events.
- Not claimed: a specific 4738 event was proven to be the direct result of the Guest-account PowerShell command.
- Not claimed: successful credential theft or account compromise was proven by this dataset view.

Event 4104 exposed the executed script

ScriptBlockText gave direct visibility into the PowerShell activity.

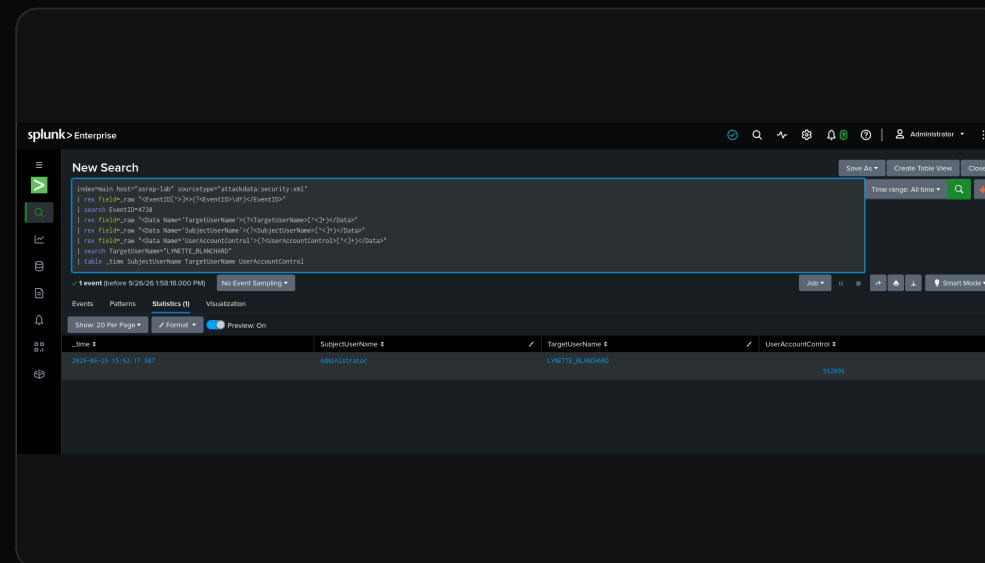
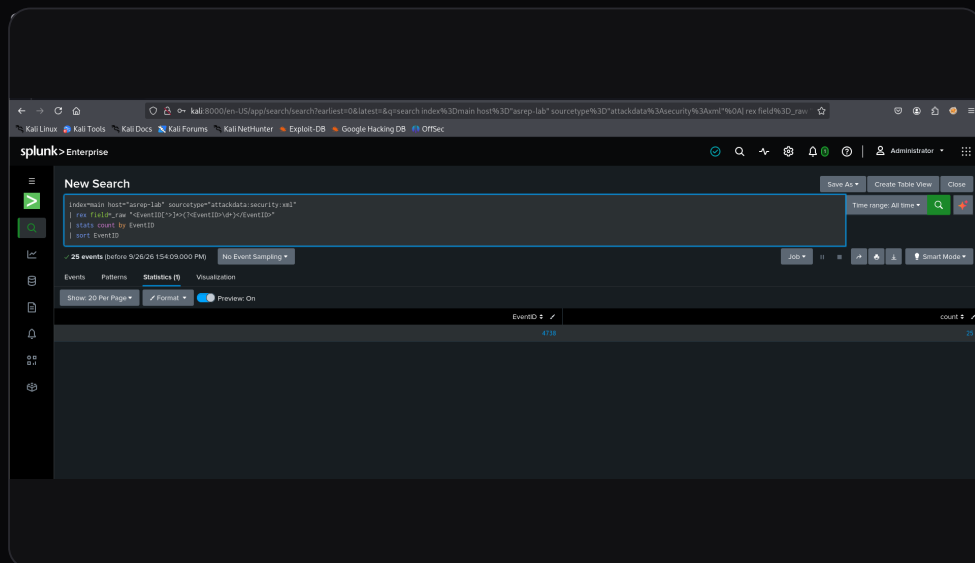
_time ▾	ScriptBlockText ▾
2022-03-28 11:07:42.525	Get-LocalUser Out-File -FilePath .\localUser.txt; Get-ADUserResultantPasswordPolicy -Identity Administrator Out-File -FilePath .\PasswordPolicy.txt; Get-ADuser Guest Set-ADAccountControl -DoesNotRequirePreAuth:\$true; Get-ADDefaultDomainPasswordPolicy Out-File -FilePath .\ADDefaultPassPolicy.txt;Enter-PSSession -ComputerName ar-win-dc-default-attack-range;[System.Environment]::UserName Out-File -FilePath .\CurrentactiveUser.txt;\$env:UserName Out-File -FilePath .\CurrentactiveUser.txt -Append;[System.Security.Principal.WindowsIdentity]::GetCurrent() Out-File -FilePath .\CurrentUserObject.txt;get-wmiobject win32_group Out-File -FilePath .\DomainGroup.txt;\$o=[activator]::CreateInstance([Type]::GetTypeFromCLSID("{9BA05972-F6A8-11CF-A442-00A0C90A8F39}")); \$item = \$o.Item() ; \$item.Document.Application.ShellExecute("cmd.exe","/c calc.exe","C:\windows\system32",\$null,0); [activator]::CreateInstance([Type]::GetTypeFromCLSID("{C08AFD90-F2A1-11D1-8455-00A0C91F3880","10.0.1.16"})).Document.Application.ShellExecute("cmd.exe","/c calc.exe","C:\windows\system32",\$null,0); Get-AdComputer -Filter * Out-File -FilePath .\AdComputer.txt;[Ref].Assembly.GetType('System.Management.Automation.AmsiUtils').GetField('amsiInitFailed','NonPublic,Static').SetValue(\$null,\$true);Invoke-Command -ComputerName ar-win-dc-default-attack-range -ScriptBlock {ipconfig};Get-WmiObject -Query â€œSELECT * FROM AntiSpywareProductâ€;Get-WmiObject -Query â€œSELECT * FROM AntiVirusProductâ€;get-wmiobject Win32_ComputerSystemProduct
2022-03-28 10:41:15.966	Get-LocalUser Out-File -FilePath .\localUser.txt Get-ADUserResultantPasswordPolicy -Identity Administrator Out-File -FilePath .\PasswordPolicy.txt Get-ADuser Guest Set-ADAccountControl -DoesNotRequirePreAuth:\$true Get-ADDefaultDomainPasswordPolicy Out-File -FilePath .\ADDefaultPassPolicy.txt Enter-PSSession -ComputerName ar-win-dc-default-attack-range [System.Environment]::UserName Out-File -FilePath .\CurrentactiveUser.txt \$env:UserName Out-File -FilePath .\CurrentactiveUser.txt -Append [System.Security.Principal.WindowsIdentity]::GetCurrent() Out-File -FilePath .\CurrentUserObject.txt get-wmiobject win32_group Out-File -FilePath .\DomainGroup.txt \$o= [activator]::CreateInstance([Type]::GetTypeFromCLSID("{9BA05972-F6A8-11CF-A442-00A0C90A8F39}")) \$item = \$o.Item() \$item.Document.Application.ShellExecute("cmd.exe","/c calc.exe","C:\windows\system32",\$null,0) [activator]::CreateInstance([Type]::GetTypeFromCLSID("{C08AFD90-F2A1-11D1-8455-00A0C91F3880","10.0.1.16"})).Document.Application.ShellExecute("cmd.exe","/c calc.exe","C:\windows\system32",\$null,0) powershell.exe Get-AdComputer -Filter * Out-File -FilePath .\AdComputer.txt [Ref].Assembly.GetType('System.Management.Automation.AmsiUtils').GetField('amsiInitFailed','NonPublic,Static').SetValue(\$null,\$true) Invoke-Command -ComouterName ar-win-dc-default-tcontreras-attack-range -ScriptBlock {ioconfig}

KEY COMMAND

```
Get-ADUser Guest | Set-ADAccountControl -DoesNotRequirePreAuth:$true
```

Plain meaning: the script selects the Guest domain account and sets it so Kerberos pre-authentication is not required.

Event 4738 showed account changes



What the security data confirms

- All 25 imported Security events in this dataset are Event ID 4738: A user account was changed.
- One filtered row shows Administrator modifying LYNETTE_BLANCHARD with a UserAccountControl value present.
- This demonstrates account-control changes in the same training dataset, but it does not prove that row is the direct effect of the Guest command shown in the PowerShell log.

A focused SPL search was built

The detection looks for PowerShell script blocks that modify the pre-authentication requirement.

SPL DETECTION

```
index=main host="asrep-lab" sourcetype="attackdata:powershell:xml"
| rex field=_raw "(?s)<Data Name='ScriptBlockText'>(?(ScriptBlockText>.*?)</Data>"
| search ScriptBlockText="*DoesNotRequirePreAuth*"
| table _time ScriptBlockText
```

splunk>Enterprise

New Search [Save As] [Create Table View] [Close]

Time range: All time [Q] [🔥]

✓ 2 events (before 9/26/26 2:03:41.000 PM) [No Event Sampling]

Job [Job] [II] [🔍] [📄] [📥] [📤] [Smart Mode]

Events Patterns **Statistics (2)** Visualization

Show: 20 Per Page [Format] [Preview: On]

_time	ScriptBlockText
2022-03-28 11:07:42.525	Get-LocalUser Out-File -FilePath .\localUser.txt; Get-ADUserResultantPasswordPolicy -Identity Administrator Out-File -FilePath .\PasswordPolicy.txt; Get-ADuser Guest Set-ADAccountControl -DoesNotRequirePreAuth:\$true; Get-ADDefaultDomainPasswordPolicy Out-File -FilePath .\ADDefaultPassPolicy.txt; Enter-PSsession -ComputerName ar-win-dc-default-attack-range; [System.Environment]::UserName Out-File -FilePath .\CurrentActiveUser.txt; \$env:UserName Out-File -FilePath .\CurrentActiveUser.txt -Append; [System.Security.Principal.WindowsIdentity]::GetCurrent() Out-File -FilePath .\CurrentUserObject.txt; get-wmiobject win32_group Out-File -FilePath .\DomainGroup.txt; \$o= [activator]::CreateInstance([type]::GetTypeFromCLSID("{9BA05972-F6A8-11CF-A442-00A0C90A8F39}")); \$item = \$o.Item(); \$item.Document.Application.ShellExecute("cmd.exe", "/c calc.exe", "C:\windows\system32", \$null, 0); [activator]::CreateInstance([type]::GetTypeFromCLSID("{C08AFD90-F2A1-11D1-8455-00A0C91F3880", "10.0.1.16")).Document.Application.ShellExecute("cmd.exe", "/c calc.exe", "C:\windows\system32", \$null, 0); Get-AdComputer -Filter * Out-File -FilePath .\AdComputer.txt; [Ref].Assembly.GetType('System.Management.Automation.AmsiUtils').GetField('amsiInitFailed', 'NonPublic, Static').SetValue(\$null, \$true); Invoke-Command -ComputerName ar-win-dc-default-attack-range -ScriptBlock (ipconfig); Get-WmiObject -Query 'SELECT * FROM AntiSpywareProduct'; Get-WmiObject -Query 'SELECT * FROM AntiVirusProduct'; get-wmiobject Win32_ComputerSystemProduct
2022-03-28 10:41:15.966	Get-LocalUser Out-File -FilePath .\localUser.txt Get-ADUserResultantPasswordPolicy -Identity Administrator Out-File -FilePath .\PasswordPolicy.txt Get-ADuser Guest Set-ADAccountControl -DoesNotRequirePreAuth:\$true Get-ADDefaultDomainPasswordPolicy Out-File -FilePath .\ADDefaultPassPolicy.txt Enter-PSsession -ComputerName ar-win-dc-default-attack-range [System.Environment]::UserName Out-File -FilePath .\CurrentActiveUser.txt \$env:UserName Out-File -FilePath .\CurrentActiveUser.txt -Append [System.Security.Principal.WindowsIdentity]::GetCurrent() Out-File -FilePath .\CurrentUserObject.txt get-wmiobject win32_group Out-File -FilePath .\DomainGroup.txt \$o= [activator]::CreateInstance([type]::GetTypeFromCLSID("{9BA05972-F6A8-11CF-A442-00A0C90A8F39}"))

to direct input to this VM, move the mouse pointer inside or press Ctrl+G

What each part does

Only the commands used in this case study.

index / host / sourcetype

Limits the search to the imported AS-REP PowerShell data.

rex

Extracts ScriptBlockText from the raw XML so it can be searched as a field.

search

Keeps events containing DoesNotRequirePreAuth.

table

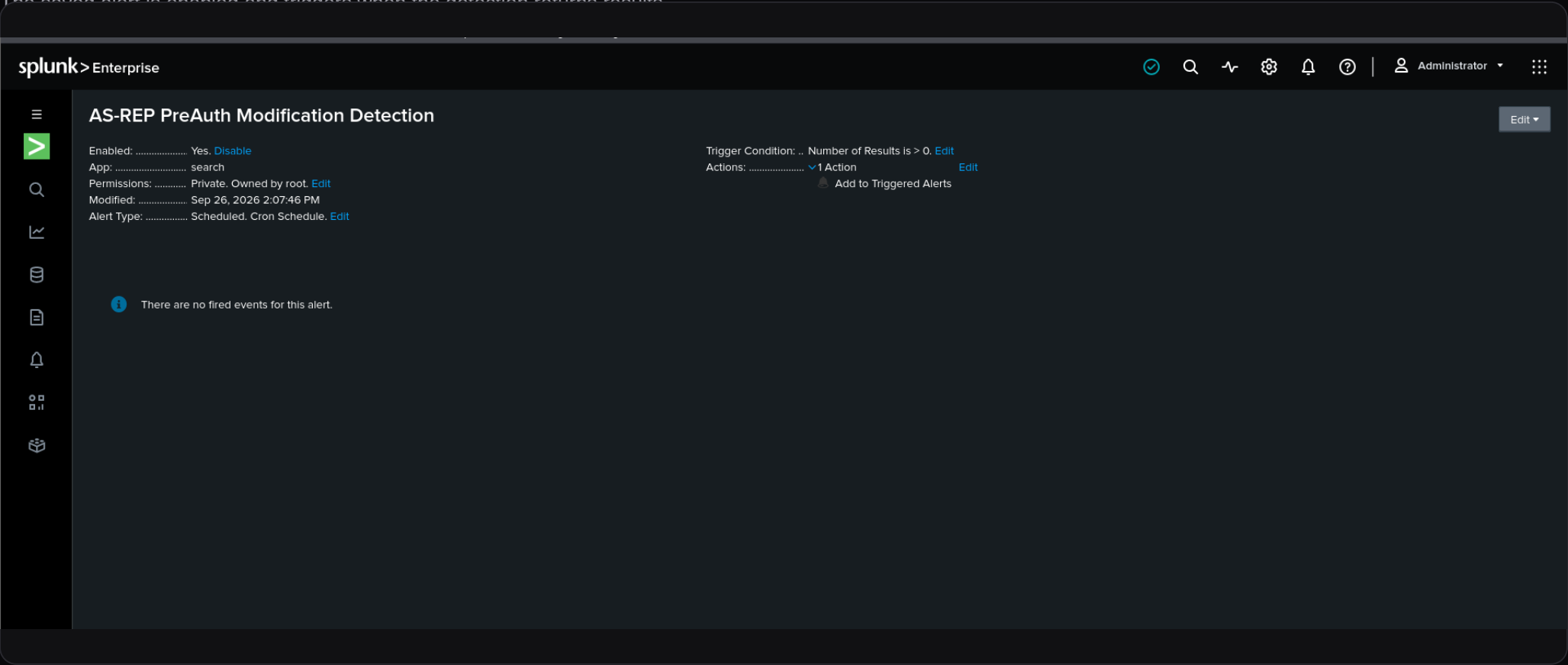
Shows only the timestamp and script text for review.

Result

The search returned 2 PowerShell events containing the pre-authentication modification string. That result became the alert condition.

Detection operationalized as a scheduled alert

The saved alert is enabled and triggers when the detection returns results



STATUS: ENABLED AS-REP PreAuth Modification Detection - scheduled search - trigger condition: Number of Results > 0 - action: Add to Triggered Alerts.

Conclusion

A concise record of what this investigation demonstrated.

Confirmed findings

- PowerShell Event 4104 exposed script content that changed Kerberos pre-authentication behavior.
- The command targeted the Guest account with `-DoesNotRequirePreAuth:$true`.
- Windows Security Event 4738 records show user-account changes in the imported dataset.
- A Splunk detection successfully identified the risky PowerShell pattern and was saved as an alert.

Limits kept explicit

- The reviewed Security row for LYNETTE_BLANCHARD was not proven to be the direct result of the Guest PowerShell command.
- This analysis does not claim that a password hash was cracked or that an account was compromised.
- The alert detects the configuration-change pattern; it is not proof of a completed AS-REP roasting attack by itself.

OFFICIAL REFERENCES

Splunk Attack Data - T1558.004 PowerShell dataset: research.splunk.com/attack_data/1f409a75-25d7-4ac6-99d1-27ae0bca0f4a/

Microsoft Learn - Set-ADAccountControl / DoesNotRequirePreAuth

Microsoft Learn - Event 4738: A user account was changed

Overall: this is a training case study based on an external dataset. The value of the work is the Splunk workflow: ingesting logs, extracting fields, reviewing evidence, building a focused detection, and documenting the limits of the conclusion.